The Poisson Distribution

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Conditions for a Poisson distribution**

State the key conditions required for a random variable to be modeled by a Poisson distribution.

Events occur independently of one another; events occur at a constant average rate over the given interval (of time or space); two events cannot occur at exactly the same instant; the number of events in non-overlapping intervals is independent.

2 **[Foundational] • Target: Computing Poisson probabilities**

A call center receives an average of 4 calls per minute, modeled by a Poisson distribution with $\lambda = 4$. Find $P(X = 6)$, using $P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$, to three decimal places.

$P(X = 6) = e^{-4}4^6/6! = (0.0183)(4096)/720 \approx 74.98/720 \approx 0.104$.

3 **[Foundational] • Target: Computing Poisson probabilities**

Using the same call center ($\lambda = 4$ calls per minute), find the probability of receiving at most 2 calls in a given minute.

$P(X \le 2) = P(X = 0) + P(X = 1) + P(X = 2)$. $P(X = 0) = e^{-4} \approx 0.0183$. $P(X = 1) = 4e^{-4} \approx 0.0733$. $P(X = 2) = 16e^{-4}/2 = 8e^{-4} \approx 0.1465$. Sum $\approx 0.0183 + 0.0733 + 0.1465 \approx 0.238$.

4 **[Foundational] • Target: Mean and variance of a Poisson distribution**

State the mean and variance of a Poisson distribution with parameter λ, and explain what is distinctive about this relationship compared to other common distributions.

For a Poisson distribution, both the mean and the variance equal λ: $E(X) = \lambda$ and $\text{Var}(X) = \lambda$. This is distinctive because for most other distributions, the mean and variance are independent parameters (e.g. the normal distribution has separate μ and σ^2) — for Poisson, a single parameter determines both.

5 **[Foundational] • Target: Sums of independent Poisson variables**

Two independent Poisson processes have rates $\lambda_1 = 3$ and $\lambda_2 = 5$ events per hour. If $X_1 \sim \text{Po}(3)$ and $X_2 \sim \text{Po}(5)$, state the distribution of $X_1 + X_2$, and find its mean and variance.

The sum of independent Poisson random variables is itself Poisson: $X_1 + X_2 \sim \text{Po}(3 + 5) = \text{Po}(8)$. Mean $= 8$; variance $= 8$ (since for Poisson, mean and variance are equal).

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6 [Foundational] • Target: Poisson as an approximation to the binomial

Explain the conditions under which a binomial distribution $X \sim B(n, p)$ can be well-approximated by a Poisson distribution, and state the appropriate value of λ to use.

The Poisson approximation to the binomial is appropriate when n is large and p is small (a common guideline: $n > 50$ and $np < 5$), so that events are rare relative to the number of trials. In that case, X is approximately $\text{Po}(\lambda)$ with $\lambda = np$.

7 [Foundational] • Target: Poisson as an approximation to the binomial

A factory produces 2000 items, each independently defective with probability 0.002. Using the Poisson approximation, find the probability that exactly 3 items are defective, to three decimal places.

$\lambda = np = 2000(0.002) = 4$. $P(X = 3) = e^{-4}4^3/3! = (0.0183)(64)/6 \approx 1.171/6 \approx 0.195$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Using cumulative Poisson probabilities

A shop receives customers at an average rate of $\lambda = 5$ per 10-minute interval. Find the probability of receiving more than 2 customers in a given 10-minute interval, given $P(X = 0) \approx 0.0067$, $P(X = 1) \approx 0.0337$, and $P(X = 2) \approx 0.0842$.

$P(X > 2) = 1 - P(X \leq 2)$. Summing the three given probabilities:
$P(X \leq 2) = 0.0067 + 0.0337 + 0.0842 = 0.1246$. So $P(X > 2) = 1 - 0.1246 = 0.8754$.

9 [Intermediate] • Target: Finding an unknown parameter from a given probability

A random variable $X \sim \text{Po}(\lambda)$ satisfies $P(X = 0) = 0.1353$. Find λ, to the nearest whole number, using $P(X = 0) = e^{-\lambda}$.

$e^{-\lambda} = 0.1353$. Taking natural logs: $-\lambda = \ln(0.1353) \approx -2.0$, so $\lambda \approx 2$.

10 [Intermediate] • Target: Variance-based reasoning with the Poisson distribution

A random variable $X \sim \text{Po}(\lambda)$ has standard deviation 3. Find λ, and then find $P(X = \lambda)$, to three decimal places.

Since variance $= \lambda$ and standard deviation $= \sqrt{\lambda} = 3$, we get $\lambda = 9$.

$P(X = 9) = e^{-9}9^9/9! = (0.0001234)(387420489)/362880 \approx 47790/362880 \approx 0.132$.

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11 **[Intermediate]** • **Target: Combining Poisson reasoning with real-world decisions**

A website server crashes on average $\lambda = 0.5$ times per week. The IT team wants to know the probability of at least one crash in a given week, given $P(X = 0) \approx 0.6065$. Should the team be concerned if their target is to keep this probability below 30%?

$P(X \geq 1) = 1 - P(X = 0) = 1 - 0.6065 = 0.3935$, or about 39.35%. Since this exceeds their target threshold of 30%, the team should be concerned — there is roughly a 39% chance of at least one crash in a given week, above their comfort threshold, suggesting they should investigate ways to reduce the average crash rate λ.

12 **[Intermediate]** • **Target: Combining a Poisson approximation with conditional probability**

Machine faults occur at an average rate of $\lambda = 1.5$ per day. Find the probability that exactly one fault occurs today, given that at least one fault occurs today, to three decimal places, using $P(X = 0) \approx 0.2231$ and $P(X = 1) \approx 0.3347$.

$P(X \geq 1) = 1 - P(X = 0) = 1 - 0.2231 = 0.7769$. Conditional probability:

$P(X = 1 \mid X \geq 1) = P(X = 1)/P(X \geq 1) = 0.3347/0.7769 \approx 0.431$.

13 **[Intermediate]** • **Target: Combining Poisson processes over different time intervals**

A radioactive source emits particles at a rate of $\lambda = 2$ per second, modeled by a Poisson distribution. Find $P(X = 3)$ for a 3-second interval, to three decimal places, using the fact that the parameter scales with the length of the interval.

For a 3-second interval, the parameter becomes $\lambda = 2 \times 3 = 6$.

$P(X = 3) = e^{-6}6^3/3! = (0.002479)(216)/6 \approx 0.5354/6 \approx 0.089$.

14 **[Intermediate]** • **Target: Combining Poisson processes over different time intervals**

Explain why the Poisson distribution is not an appropriate model for the number of heads obtained when flipping a fair coin 10 times, referencing the conditions required for a Poisson model.

The number of heads in 10 coin flips is naturally modeled by a binomial distribution $B(10, 0.5)$, not a Poisson distribution, because a Poisson model requires events to be rare (a small probability of occurrence per trial, with np moderate while n is large) — here $p = 0.5$ is far too large for the Poisson approximation to be valid, and there is a fixed, small number of trials ($n = 10$) rather than events occurring continuously over time or space.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Comparing two independent Poisson processes • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Two independent Poisson processes have rates $\lambda_A = 3$ and $\lambda_B = 2$ per hour. Find the probability that, in a given hour, process A produces exactly 2 events and process B produces exactly 1 event, to three decimal places, given $P(X_A = 2) \approx 0.2240$ and $P(X_B = 1) \approx 0.2707$.

Since the two processes are independent, the joint probability is the product of the individual probabilities: $P(X_A = 2) \times P(X_B = 1) = 0.2240 \times 0.2707 \approx 0.0606$.

16 [Challenging] • Target: Rescaling the Poisson parameter for a different interval • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

A radio station receives an average of 12 requests per hour. Find the appropriate Poisson parameter λ to use for a 15-minute segment.

Since the rate is 12 per hour and 15 minutes is a quarter of an hour, the parameter for that segment is $\lambda = 12/4 = 3$.

17 [Challenging] • Target: The mode of a Poisson distribution • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

A random variable $X \sim \text{Po}(4)$ has an integer parameter $\lambda = 4$. State the mode(s) of X (the most likely value or values), and explain why a Poisson distribution with an integer λ has two modes, using the ratio $P(X = k)/P(X = k - 1) = \lambda/k$.

At $k = \lambda$, the ratio $P(X = k)/P(X = k - 1) = \lambda/k = \lambda/\lambda = 1$, so $P(X = \lambda) = P(X = \lambda - 1)$ exactly. This means both $X = 3$ and $X = 4$ share the same (maximum) probability, so the distribution has two modes: 3 and 4. This only happens when λ is a positive integer; for a non-integer λ, there is a single mode at $\lfloor \lambda \rfloor$.

18 [Challenging] • Target: Assessing Poisson plausibility from sample data • Cross-Lesson: 5.01 Further Integration Techniques

A biologist counts the number of bacteria colonies in 10 equal-sized petri dishes: $3, 5, 4, 6, 5, 4, 3, 5, 6, 4$. Find the sample mean and the unbiased sample variance, and comment on whether a Poisson model seems plausible for this data.

Mean $= (3 + 5 + 4 + 6 + 5 + 4 + 3 + 5 + 6 + 4)/10 = 45/10 = 4.5$. Squared deviations from the mean sum to $4(2.25) + 6(0.25) = 9 + 1.5 = 10.5$ (the values 3 and 6 each deviate by 1.5, contributing 2.25 each, four times in total; the values 4 and 5 each deviate by 0.5, contributing 0.25 each, six times in total). Unbiased sample variance: $10.5/(10 - 1) = 10.5/9 \approx 1.167$. Since the sample variance (≈ 1.17) is much smaller than the sample mean (4.5), a Poisson model seems implausible for this data — a genuine Poisson dist.

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19 [Challenging] • Target: Changing Time Intervals & Sum of Independent Poisson RVs • Cross-Lesson: 5.02 Differential Equations

If $X \sim B(150, \frac{1}{100})$, use a Poisson approximation to find $P(X = 1)$, correct to 4 decimal places.

Final Answer: 0.3347. $\lambda = np = 150 \times \frac{1}{100} = 1.5$. $P(X = 1) \approx \frac{e^{-1.5}1.5^1}{1!} \approx 0.3347$

20 [Challenging] • Target: Poisson Approximation to the Binomial Distribution • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

If $X \sim B(300, \frac{1}{100})$, use a Poisson approximation to find $P(X = 1)$, correct to 4 decimal places.

Final Answer: 0.1494. $\lambda = np = 300 \times \frac{1}{100} = 3.0$. $P(X = 1) \approx \frac{e^{-3.0}3.0^1}{1!} \approx 0.1494$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Identifying an appropriate Poisson scenario

Which of the following scenarios is best modeled by a Poisson distribution?

a. The number of typing errors on a randomly selected page of a manuscript. ✓

b. The number of heads in 20 coin tosses.

c. The height of a randomly selected adult.

d. The number of correct answers on a 10-question multiple-choice test, each with 4 options.

Typing errors occur rarely and independently across a fixed amount of text (a page), matching the Poisson conditions of a constant average rate with no fixed number of trials. The coin-toss and multiple-choice scenarios are binomial (a fixed number of trials, each with the same success probability), and height is a continuous variable following approximately a normal distribution — neither fits the Poisson framework.

22 [Foundational] • Target: The Poisson Probability Mass Function

If $X \sim \text{Po}(\lambda)$, what is the probability mass function $P(X = k)$ for $k = 0, 1, 2, \ldots$?

a. $P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$ ✓

b. $P(X = k) = \frac{\lambda^k e^{\lambda}}{k!}$

c. $P(X = k) = \binom{n}{k}\lambda^k(1 - \lambda)^{n-k}$

d. $P(X = k) = \lambda e^{-\lambda k}$

The Poisson probability mass function is given by $P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$ for $k \in \{0, 1, 2, \ldots\}$.

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23 [Foundational] • Target: The Poisson Probability Mass Function

A call center receives an average of 3 calls per minute ($X \sim \text{Po}(3)$). What is the probability of receiving exactly 2 calls in a given minute?

a. $\frac{9}{2}e^{-3} \approx 0.224$ ✓

b. $3e^{-3}$

c. $\frac{27}{6}e^{-3}$

d. e^{-3}

$P(X = 2) = \frac{e^{-3}3^2}{2!} = \frac{9}{2}e^{-3} = 4.5e^{-3} \approx 0.2240$.

24 [Foundational] • Target: The Poisson Probability Mass Function

For $X \sim \text{Po}(\lambda)$, what is $P(X = 0)$?

a. $e^{-\lambda}$ ✓

b. 0

c. 1

d. $\lambda e^{-\lambda}$

Substituting $k = 0$: $P(X = 0) = \frac{e^{-\lambda}\lambda^0}{0!} = \frac{e^{-\lambda} \cdot 1}{1} = e^{-\lambda}$.

25 [Foundational] • Target: Mean, Variance, and Poisson Characteristics

What is a distinctive mathematical property of any Poisson random variable $X \sim \text{Po}(\lambda)$?

a. **The mean and variance are strictly equal:** $E(X) = \text{Var}(X) = \lambda$ ✓

b. The mean is always λ^2

c. The standard deviation equals λ

d. The median always equals $\lambda + 1$

A defining feature of the Poisson distribution is that its expected value and variance are both equal to the rate parameter λ.

26 [Foundational] • Target: Mean, Variance, and Poisson Characteristics

If a Poisson random variable has variance $\text{Var}(X) = 16$, what is its standard deviation?

a. 4 ✓

b. 16

c. 256

d. $\sqrt{4} = 2$

Standard deviation is $\sigma = \sqrt{\text{Var}(X)} = \sqrt{16} = 4$.

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27 [Foundational] • Target: Mean, Variance, and Poisson Characteristics

Under what practical conditions does a Poisson process serve as a valid model?

a. Events occur independently, singly, and at a constant average rate across continuous time or space ✓

b. There are exactly two trials with equal probability

c. Events must occur in clusters at predetermined times

d. The probability of an event increases as time elapses

A Poisson process assumes events happen independently at a constant average rate, without simultaneous multiple arrivals.

28 [Foundational] • Target: Changing Time Intervals & Sum of Independent Poisson RVs

Flaws on a wire occur at an average rate of 1.5 per meter. What is the distribution of flaws Y in a 4-meter segment of wire?

a. $Y \sim \text{Po}(6)$ ✓

b. $Y \sim \text{Po}(1.5)$

c. $Y \sim \text{Po}(4)$

d. $Y \sim \text{Po}(9)$

Poisson rates scale linearly with interval size: $\lambda_{\text{new}} = 1.5 \times 4 = 6$. Thus $Y \sim \text{Po}(6)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Changing Time Intervals & Sum of Independent Poisson RVs

If $X \sim \text{Po}(4)$ and $Y \sim \text{Po}(7)$ are independent random variables, what is the distribution of $W = X + Y$?

a. $W \sim \text{Po}(11)$ ✓

b. $W \sim \text{Po}(28)$

c. $W \sim \text{Po}(3)$

d. $W \sim N(11, 28)$

The sum of independent Poisson random variables is itself a Poisson random variable whose parameter is the sum of the individual rates: $\lambda_W = 4 + 7 = 11$.

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30 [Intermediate] • Target: Changing Time Intervals & Sum of Independent Poisson RVs

If $W \sim \text{Po}(11)$, what is the variance of W?

a. 11 ✓

b. 121

c. $\sqrt{11}$

d. 22

For any Poisson random variable $W \sim \text{Po}(\lambda)$, $\text{Var}(W) = \lambda = 11$.

31 [Intermediate] • Target: Poisson Approximation to the Binomial Distribution

When can a Binomial distribution $B(n, p)$ be well approximated by a Poisson distribution $\text{Po}(\lambda)$?

a. **When n is very large ($n > 50$) and p is very small ($p < 0.1$), with $\lambda = np$** ✓

b. When n is small and $p = 0.5$

c. When $np \geq 10$ and $n(1-p) \geq 10$

d. For any binomial distribution where $n > 30$

The Poisson distribution is the limiting form of the Binomial distribution as $n \to \infty$ and $p \to 0$ such that $np = \lambda$ remains moderate.

32 [Intermediate] • Target: Poisson Approximation to the Binomial Distribution

A factory produces microchips with defect probability $p = 0.002$. In a batch of $n = 1000$ chips, approximate the probability of finding zero defects.

a. $e^{-2} \approx 0.135$ ✓

b. $e^{-0.002}$

c. $1 - e^{-2}$

d. 0.002

$\lambda = np = 1000(0.002) = 2$. $P(X = 0) \approx e^{-\lambda} = e^{-2} \approx 0.1353$.

33 [Intermediate] • Target: Poisson Approximation to the Binomial Distribution

If $X \sim \text{Po}(\lambda)$ has $P(X = 1) = P(X = 2)$, find the value of λ.

a. $\lambda = 2$ ✓

b. $\lambda = 1$

c. $\lambda = 0.5$

d. $\lambda = 4$

$\frac{e^{-\lambda}\lambda^1}{1!} = \frac{e^{-\lambda}\lambda^2}{2!} \Rightarrow \lambda = \frac{\lambda^2}{2} \Rightarrow 1 = \frac{\lambda}{2} \Rightarrow \lambda = 2$.

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34 [Intermediate] • Target: The mode of a Poisson distribution

If $X \sim B(300, \frac{1}{100})$, use a Poisson approximation to find $P(X = 3)$, correct to 4 decimal places.

a. 0.2640

b. 0.1680

c. 0.1008

d. **0.2240** ✓

$\lambda = np = 300 \times \frac{1}{100} = 3.0$. $P(X = 3) \approx \frac{e^{-3.0}3.0^3}{3!} \approx 0.2240$

35 [Intermediate] • Target: Identifying an appropriate Poisson scenario

If $X \sim B(200, \frac{1}{100})$, use a Poisson approximation to find $P(X = 0)$, correct to 4 decimal places.

a. 0.1753

b. 0.2707

c. **0.1353** ✓

d. 0.2153

$\lambda = np = 200 \times \frac{1}{100} = 2.0$. $P(X = 0) \approx \frac{e^{-2.0}2.0^0}{0!} \approx 0.1353$

36 [Intermediate] • Target: Assessing Poisson plausibility from sample data

If $X \sim B(150, \frac{1}{100})$, use a Poisson approximation to find $P(X = 3)$, correct to 4 decimal places.

a. 0.0471

b. **0.1255** ✓

c. 0.0141

d. 0.2510

$\lambda = np = 150 \times \frac{1}{100} = 1.5$. $P(X = 3) \approx \frac{e^{-1.5}1.5^3}{3!} \approx 0.1255$

37 [Intermediate] • Target: The Poisson Probability Mass Function

If $X \sim B(150, \frac{1}{25})$, use a Poisson approximation to find $P(X = 3)$, correct to 4 decimal places.

a. 0.0446

b. **0.0892** ✓

c. 0.1606

d. 0.1339

$\lambda = np = 150 \times \frac{1}{25} = 6.0$. $P(X = 3) \approx \frac{e^{-6.0}6.0^3}{3!} \approx 0.0892$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Mean, Variance, and Poisson Characteristics • Cross-Lesson: 5.01 Further Integration Techniques

If $X \sim B(200, \frac{1}{50})$, use a Poisson approximation to find $P(X = 1)$, correct to 4 decimal places.

a. 0.1954

b. 0.0183

c. **0.0733** ✓

d. 0.1465

$\lambda = np = 200 \times \frac{1}{50} = 4.0$. $P(X = 1) \approx \frac{e^{-4.0}4.0^1}{1!} \approx 0.0733$

39 [Challenging] • Target: Changing Time Intervals & Sum of Independent Poisson RVs • Cross-Lesson: 5.02 Differential Equations

If $X \sim B(300, \frac{1}{50})$, use a Poisson approximation to find $P(X = 3)$, correct to 4 decimal places.

a. 0.1339

b. 0.0446

c. 0.1606

d. **0.0892** ✓

$\lambda = np = 300 \times \frac{1}{50} = 6.0$. $P(X = 3) \approx \frac{e^{-6.0}6.0^3}{3!} \approx 0.0892$

40 [Challenging] • Target: Poisson Approximation to the Binomial Distribution • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

If $X \sim B(300, \frac{1}{50})$, use a Poisson approximation to find $P(X = 0)$, correct to 4 decimal places.

a. 0.0425

b. 0.0446

c. **0.0025** ✓

d. 0.0149

$\lambda = np = 300 \times \frac{1}{50} = 6.0$. $P(X = 0) \approx \frac{e^{-6.0}6.0^0}{0!} \approx 0.0025$

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41 [Challenging] • Target: Conditions for a Poisson distribution • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

A school offers two optional subjects: Art and Music. The number of students enrolled in each subject and the number of students enrolled in both is shown in the Venn diagram below. The total number of students in the school is 120. Let A be the set of students enrolled in Art and M be the set of students enrolled in Music. Given that $n(A \cup M) = 90$, find the number of students enrolled in neither Art nor Music.

a. 20

b. **30** ✓

c. 40

d. 10

The Venn diagram shows that $n(A \cap M) = 20$, $n(A \setminus M) = 3x$, and $n(M \setminus A) = 4x$. The total number of students in the school is 120. We are given that $n(A \cup M) = 90$. The formula for the union of two sets is $n(A \cup M) = n(A \setminus M) + n(M \setminus A) + n(A \cap M)$. Substituting the given values: $90 = 3x + 4x + 20$. Subtracting 20 from both sides: $70 = 7x$. Dividing by 7: $x = 10$. Now we can find the number of students in Art only and Music only.

42 [Challenging] • Target: Computing Poisson probabilities • Cross-Lesson: 6.01 Matrices (AI HL)

X takes values $0, 1, 2$ with probabilities $\frac{1}{5}, \frac{3}{10}, \frac{1}{2}$. Find $\text{Var}(X)$.

a. $\frac{61}{100}$ ✓

b. $\frac{121}{100}$

c. $\frac{73}{100}$

d. $\frac{4}{5}$

Working through this statistics step: $E(X) = \frac{13}{10},\ E(X^2) = \frac{23}{10} \Rightarrow \text{Var}(X) = \frac{61}{100}$

43 [Challenging] • Target: Mean and variance of a Poisson distribution • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

A discrete random variable X takes values $0, 3, 4$ with probabilities $\frac{1}{10}, \frac{3}{10}, \frac{3}{5}$ respectively. Find $E(X)$.

a. 1

b. 6

c. $\frac{33}{10}$ ✓

d. 2

Working through this statistics step: $E(X) = \sum xP(X = x) = 0(\frac{1}{10}) + 3(\frac{3}{10}) + 4(\frac{3}{5}) = \frac{33}{10}$

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44 [Challenging] • Target: Sums of independent Poisson variables • Cross-Lesson: 6.03 Markov Chains (AI HL)

A discrete random variable X takes values $0, 2, 4$ with probabilities $\frac{1}{10}, \frac{1}{5}, \frac{7}{10}$ respectively. Find $E(X)$.

a. 5

b. 0

c. $\frac{16}{5}$ ✓

d. 4

Working through this statistics step: $E(X) = \sum xP(X = x) = 0(\frac{1}{10}) + 2(\frac{1}{5}) + 4(\frac{7}{10}) = \frac{16}{5}$

45 [Challenging] • Target: Poisson as an approximation to the binomial • Cross-Lesson: 1.01 Proof by Mathematical Induction

A discrete random variable X takes values $0, 2, 4$ with probabilities $\frac{3}{10}, \frac{1}{10}, \frac{3}{5}$ respectively. Find $E(X)$.

a. 0

b. 5

c. 1

d. $\frac{13}{5}$ ✓

Working through this statistics step: $E(X) = \sum xP(X = x) = 0(\frac{3}{10}) + 2(\frac{1}{10}) + 4(\frac{3}{5}) = \frac{13}{5}$

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